

Gentamicin coated iron oxide nanoparticles as novel antibacterial agents

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Abstract

Applications of different types of magnetic nanoparticles for biomedical purposes started a long time back. The concept of surface functionalization of the iron oxide nanoparticles with antibiotics is a novel technique which paves the path for further application of these nanoparticles by virtue of their property of superparamagnetism. In this paper, we have synthesized novel iron oxide nanoparticles surface functionalized with Gentamicin. The average size of the particles, concluded from the HR-TEM images, came to be around 14 nm and 10 nm for unmodified and modified nanoparticles, respectively. The magnetization curve $M(H)$ obtained for these nanoparticles are typical of superparamagnetic nature and having almost zero values of coercivity and remanance. The release properties of the drug coated nanoparticles were studied; obtaining an S shaped profile, indicating the initial burst effect followed by gradual sustained release. In-vitro investigations against various gram positive and gram negative strains viz. *Staphylococcus aureus*, *Bacillus subtilis*, *Escherichia coli* and *Pseudomonas aeruginosa* indicated significant antibacterial efficiency of the drug-nanoparticle conjugate. The MIC values indicated that a small amount like 0.2 mg/ml of drug capped particles induce about 98% bacterial death. The novelty of the work lies in the drug capping of the nanoparticles, which retains the superparamagnetic nature of the iron oxide nanoparticles and the medical properties of the drug simultaneously, which is found to be extremely blood compatible.

Keywords: magnetic nanoparticles; co-precipitation; surface functionalization; superparamagnetism.

Motivation: Nanoparticles are widely in research due to its tunable properties and diverse applications. Antibacterial activities of various nanoparticles have been reported so far. Merely with the spread of more complex and resistant bacterial strains giving rise to different infectious diseases demanded the modification of nanoparticles for more dependable functioning. Drug loading on nanoparticles is a novel concept. In this work, a novel drug-nanoparticle conjugate have been developed which is way effective against different strains of bacteria. The singularity of this research is in the drug capping followed by its in-vitro tests and analysis of haemotoxicity, which is scarcely reported elsewhere.

1. Introduction:

Fighting against infections has become more challenging due to the emergence of various bacterial strains resistant to antibiotics. To overcome this, scientists are looking for novel and more efficient antibacterial agents. Many researchers have reported various kinds of nanoparticles and their antibacterial activities (1–5). Metal oxides have gained much significance in this field owing to its high efficiency and capability of being tailor made size and shape (6–11). Surface modifications (12–14) of the nanoparticles are also being done widely for better performance of the particles. Various drugs have also been incorporated along with nanoparticles to enhance its antimicrobial nature (15–21). In this paper, we have reported the synthesis of a novel iron oxide nanoparticles conjugated with a drug, Gentamicin and compared its antimicrobial property with unmodified iron oxide nanoparticles. Iron oxide nanoparticles are chosen due their property of superparamagnetism, which stops them from retaining their magnetism after the external magnetic field is removed and this property promotes these particles to act as potential drug carrying vehicles (22–24). Four kinds of gram positive and gram

negative bacteria are used in this study to demonstrate the efficacy of these nanoparticles as antibacterial.

2. Materials and Methods

2.1 Synthesis of Iron oxide nanoparticles

For preparing iron oxide nanoparticles, the technique of co-precipitation was used as reported in the literatures (25,26). The raw materials used in this technique include ferric chloride (FeCl_3), ferrous chloride (FeCl_2) and sodium hydroxide (NaOH) which were obtained from Merck Specialties, India. 30 ml of Ferrous Chloride (0.2 M) and ferric chloride (0.3M) was mixed in a 200 ml beaker. With continuous stirring, 1.5 M NaOH was added to bring the pH of the solution 12. The stirring was continued at 700 rpm to complete the reaction. The precipitate was separated from the solution by centrifuging it at 10000 rpm for 10 mins. The product was washed and then allowed to dry in a hot air oven overnight. The resulting product was calcined at 400°C for 4 hours. The dried powder was collected and characterized.

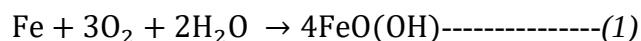
2.2 Synthesis of Gentamicin loaded iron oxide nanoparticles

The preparation of Gentamicin loaded nanoparticles was done by the method described by Palanikumar et al (27) with slight modifications. A solution of 5 gm Gentamicin in 50 mL water was added slowly to the mixture of ferrous and ferric chloride solution (in 2:3 ratios) and the resulting solution were stirred continuously for 30 mins. Sodium Hydroxide in 25ml water was introduced in the solution which resulted in the appearance of a black precipitate. The precipitate was washed twice with distilled water. It was dried in a hot air oven at 100°C for 8 hours.

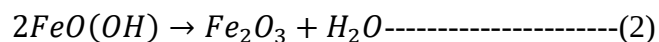
2.3 *Characterization of the nanoparticles*

Both kinds of synthesized particles were characterized for shape and size using JEM-2100 High Resolution Transmission Electron Microscope, JEOL, JAPAN with an accelerating voltage of 20KeV and camera constant of 40.1835mm. The samples were dispersed in acetone and were ultrasonicated for 30 mins, giving a 2 min gap after each 5mins of operation. The machine was operated in Image mode and well as Diffraction mode to obtain the SAED (Selected Area Electron Diffraction) images. The crystallinity of the particles were observed by Pananalytical High Resolution X-ray Diffraction system, PW 3050/60 with CuK α radiation ($\lambda = 1.5418 \text{ \AA}$) between $2\Theta = 20^\circ$ to 80° . The spectrum was recorded at a generator voltage of 40KV, tube current of 30mA and a step size of 0.016° . The chemical bonds and phases present in the nanoparticles (both raw and coated) were analyzed using a Perkin Elmer FTIR. Few milligrams of the sample were taken and it was mixed with potassium bromide (used as a binding agent) and it was hydraulically pressed to create pellets before the analysis was done. The magnetic properties of the synthesized magnetite nanoparticles were studied using an Evercool SQUID VSM DC magnetometer, manufactured by quantum Design, USA in the field range of $\pm 10\text{kOe}$ at 300K. The ZFC-FC curves were obtained at a field of 100 Oe, measured from 5K – 300K. The zeta potential of the synthesized nanoparticles was measured in a polystyrene zeta cell by Malvern Zeta-sizer ZS90 Nano-series (UK). The solutions were prepared in distilled water and the pH was adjusted by Sodium Hydroxide (NaOH) and Hydrochloric acid (HCl) solution.

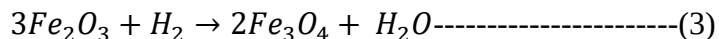
2.4 *Effect of parameters:* The shape and size of the nanoparticles and the drug loading highly depend upon the synthesis parameters such as pH, stirring speed, reagent concentration, the choice of solvent, ageing time etc. The formation of iron oxide nanoparticles can be summarized using the following reaction:



The resulting hydrated iron(III) oxide, written here as Fe(O)OH, dehydrates around 200 °C.



Partial reduction with hydrogen at about 400 °C produces magnetite, a black magnetic material that contains both Fe(III) and Fe(II).



The abundance of OH⁻ is highly required to ensure proper formation of magnetite particles. A high concentration of OH⁻ drives the proper dehydration of iron (III) oxide in step 2. Andrade et al (28) reported the effect of initial pH on formation of magnetite (Fe₃O₄) nanoparticles during co-precipitation. A significant difference in composition of resulting particles was seen by varying the initial pH. Those synthesized from pH 9.7 to 10.6 was found to contain pure magnetite with very less other forms of oxides. Other products, like maghemite and goethite appeared when the synthesis was done in pH values below 8.5. The shear stress applied to the reaction mixture by the magnitude of stirring speed is inversely proportional to the nanoparticle size(29). However, too high stirring speed is not recommended for drug loading as it will hinder the functionalization and the drug will be eluted in the reaction mixture only. It has been studied that the physical properties like interfacial tension, surface charge and viscosity of the chosen organic solvents highly affect the nanoparticle synthesis. The particle size has a direct impact on the stability, drug entrapment and release of the nanoparticle (30). Drug loading and the amount of drug associated with the carrier over time after administration are of significance in the formulation of a drug using a nanoparticle carrier. The equilibration time for the proper loading of the drug on the nanoparticles should be optimized. Longer equilibration time is not desirable because of the experimental hazards and longer reaction time may affect the nanoparticle size,

whereas shorter equilibration time affects the nucleation and growth of the particles. The release of the drugs into the solvent is an effective measure of this and it is usually done by UV spectrophotometric method.

2.5 *Release study of the drug*

The study of the release behavior of the antibiotic from the nanoparticles is of prime importance. For this, the spectrophotometric method is adopted. OPA reagent was used as a derivatizing agent as raw Gentamicin absorbs neither visible light nor UV light (31). It reacts with the amino groups of the antibiotic to yield chromophoric products. For the OPA reagent, a fresh solution of 70mg OPA was mixed with 1ml methanol and 95 ml buffer 10.5(25gm/l boric acid+ 0.3% Brij+0.2% 2-mercaptoethanol) and kept in dark for 24 hours before use. The PBS buffer was used to imitate human simulated fluid.

A scan of Gentamicin sulphate shows maximum absorbance at 330 nm. The absorbance of the solution (NP+PBS) was measured at time intervals of 24 hours.

2.6 *Bacterial Culture*

Four strains of bacteria, *S.aureus* (NCIM 2120), *E.coli* (NCIM 2137), *B.subtilis* (NCIM 2063) and *P.aeruginosa* (NCIM 5029) were obtained from National Collection of Industrial Microorganisms (NCIM), Pune, India. The bacterial cultures were grown in Luria Bertani (LB) broth procured from HiMedia, Mumbai, India. It was incubated 37°C for 24 hours under gentle stirring at 100 revolutions/min so that the bacteria become fully grown, and then it was used for further applications.

2.7 *Zone of inhibition*

If an antibacterial agent stops the bacteria from growing or kills the bacteria, there will be a zone around where the bacterial growth is not enough to be visible and this is typically called zone of

inhibition (32,33). For this, various concentrations of nanoparticles (0.1-5.0 mg/ml) were taken and circular filter paper disk was ultra-sonicated in it and kept on bacteria spread plates for 24 hour incubation. The well disk diffusion was also done by cutting a well in the agar plate and filling it with 60µl of nanoparticle solution. The diameter of the clear zone around the disk/well was measured and quantified.

2.8 Colony forming Unit Study

A very efficient method of testing the antibacterial efficacy of nanoparticles is the CFU Reduction test that is used widely (34,35). For this, various concentrations of NP + bacteria were incubated with stirring for 24 hours, serially diluted and spread evenly on agar plates. The number of colonies appearing on the plate after overnight incubation were counted and quantified in terms of CFU/ml.

2.9 Amino Acid Leakage

Any antimicrobial agent kills bacteria by damaging its nucleic acid components (DNA and RNA) by penetrating through its cell wall. Thus, the efficiency of the nanoparticles can be measured in terms of nucleic acid leakage from the bacterial cells. For this, the absorbance method was adopted. The absorbance of the NP and bacterial solution was measured at 260nm after every 1 hour. The wavelength of 260 nm was chosen because the nucleic acids show its maximum absorbance at 260nm.

2.10 Haemotoxicity

The drug loaded nanoparticles were dissolved in autoclaved water according to desired concentrations (0.2 and 2.0 mg/ml). The haemolytic assay was carried out following the protocol by reported by Dutta et. al (36) with some modifications. Briefly, blood was collected from human volunteers and the RBC was separated by centrifuging at 1500 rpm for 5 minutes. The

collected RBC pellet was diluted with Histopaque®-1077 in the ratio of 1:1. The suspension was then added to PBS buffer(negative control), 1% Triton X-100 (positive control) and the nanoparticle solutions, mixed well and incubated at 37°C for 1 hour. After centrifugation at 1200 rpm for 5 minutes, the supernatants were transferred to a 96 well plate and the absorbance was measured at 550 nm using a microplate reader (Biorad US). Haemolytic activity of the different concentrations of the conjugate was reported in relative to the percent lysis done by the positive and negative control solutions, by the following formula:

$$\text{Percent haemolysis} = \frac{\text{Absorbance of sample} - \text{absorbance of negative control}}{\text{Absorbance of positive control} - \text{absorbance of negative control}} \times 100 \quad \text{---(4)}$$

3. Results

3.1 Characterization of NPs

3.1.1 High Resolution Transmission Electron microscope

Figure 1 shows the HR-TEM image of nanoparticles. For obtaining a size distribution of the nanoparticles, a frequency histogram is plotted which is shown in figure 2.

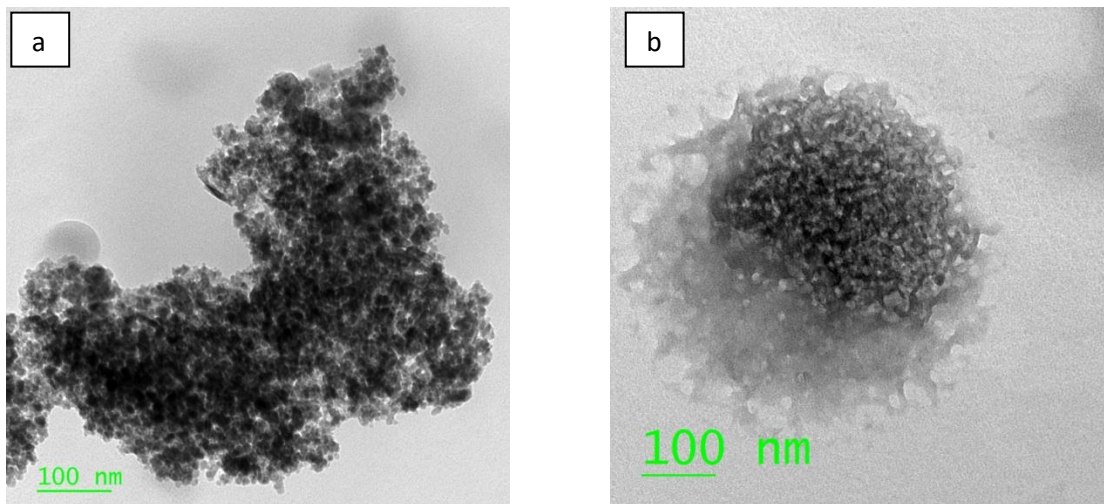


Figure 1 HRTEM image of (a) unloaded iron oxide and (b) drug loaded iron oxide nanoparticle

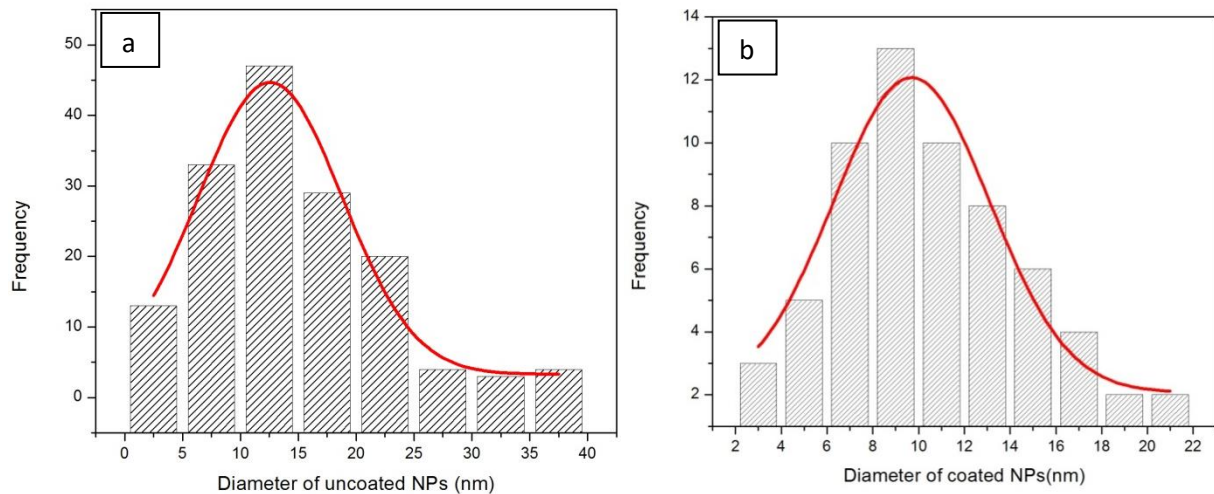


Figure 2 Size distribution histograms of (a) uncoated and (b) drug loaded iron oxide nanoparticles

From the HRTEM results, it was concluded that the average diameter of the uncoated particles is 14.21 nm and that of coated particles is 10.76 nm. The shrinkage in the size in the drug loaded nanoparticles may be attributed to the lattice strain generated due to the large Gentamicin sulfate molecules, which have inhibited the nucleation and growth of the iron oxide nanoparticles during the reaction. The HRTEM is operated in SAED mode to obtain the images which is shown in figure 3.

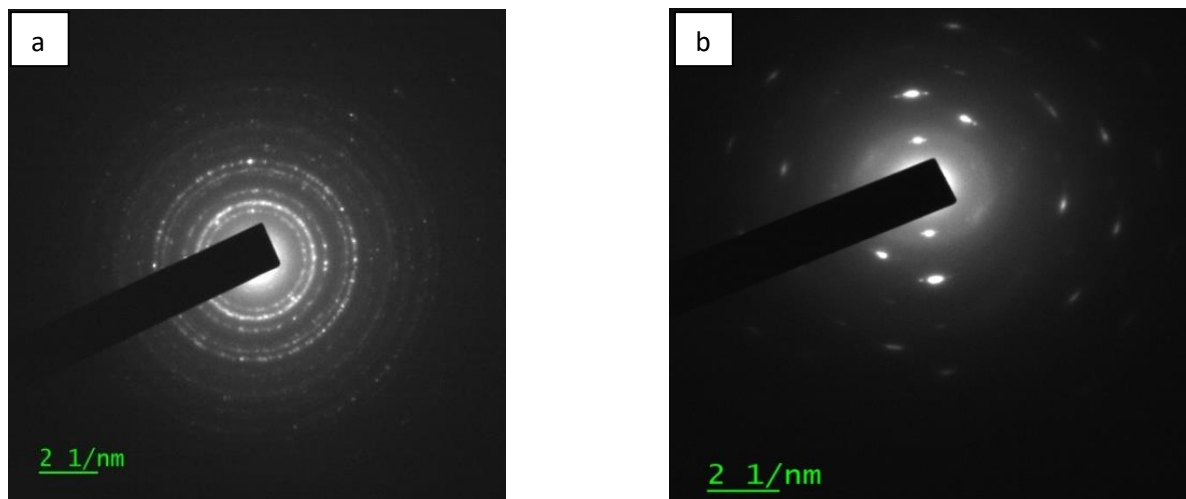


Figure 3 The SAED spectra of the (a) pristine Fe_3O_4 nanoparticles showing circles with bright spots and (b) coated Fe_3O_4 nanoparticles showing bright spots without prominent rings

The SAED image for uncoated particles show prominent rings with the appearance of bright spots embedded in the rings which implies the polycrystalline nature of the particles with lattice constant of $7.94 \text{ \AA}$. The absence of any prominent ring in figure 3 (b) indicates high crystallinity of the particles with a spherical shape with lattice constant of $5.93 \text{ \AA}$.

3.1.2 Fourier transform infrared spectroscopy

To get an insight of the functional groups present in both raw and uncoated iron oxide NPs, FTIR spectra was obtained, shown in figure 4. The different peaks in both the cases, attribute to different functional groups, tabulated in table 1. In case of uncoated particles, the peak at 3400.91 cm^{-1} is attributed to the stretching vibrations of OH^- , which is assigned to OH^- absorbed by Fe_3O_4 nanoparticles. The peak at 982.16 cm^{-1} corresponds to bending vibration associated to the O – H bond. The sample exhibits two intense peaks, in 562 and 639 cm^{-1} band, that are due to the stretching vibration mode associated to the metal-oxygen absorption band (Fe – O bonds in the crystalline lattice of Fe_3O_4). The spectra for the coated ones show the presence of both linear and cyclic ether linkages, at 1122 and 1135 cm^{-1} . The C-N stretching of Gentamicin is also visible at 1339 cm^{-1} .

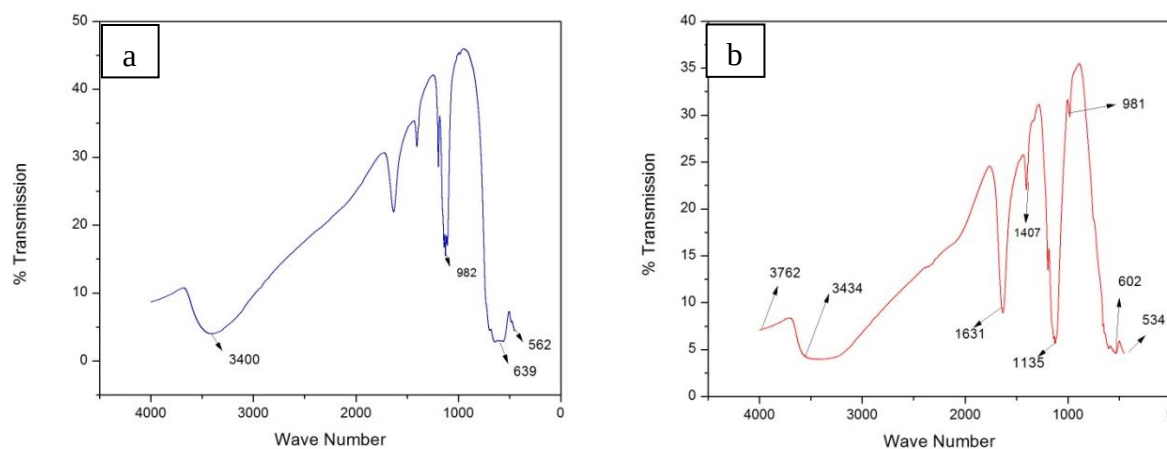


Figure 4 FTIR Spectra for (a) uncoated and (b) coated Nanoparticles

3.1.3 X-ray diffraction

The XRD spectra of both the particles are shown in figure 5. The diffraction peaks for raw iron oxide corresponds to $2\theta = 30^\circ(220)$, $35^\circ(311)$, $43^\circ(400)$, $57^\circ(511)$, $63^\circ(440)$ and $74^\circ(533)$ were observed in the image. These are in coherence with the standard JCPDS 19-0629 pattern characteristics of magnetite cubic inverse spinel structure. The sample XRD indicates the ultra-fine nature and small crystallite size of the particles. The average crystallite size of the uncoated nanoparticles obtained from the XRD spectra using Scherrer Calculator came out to be 14.95 nm and that of Gentamicin coated particles came out to be 10.37 nm which is in coherence with the HR-TEM results.

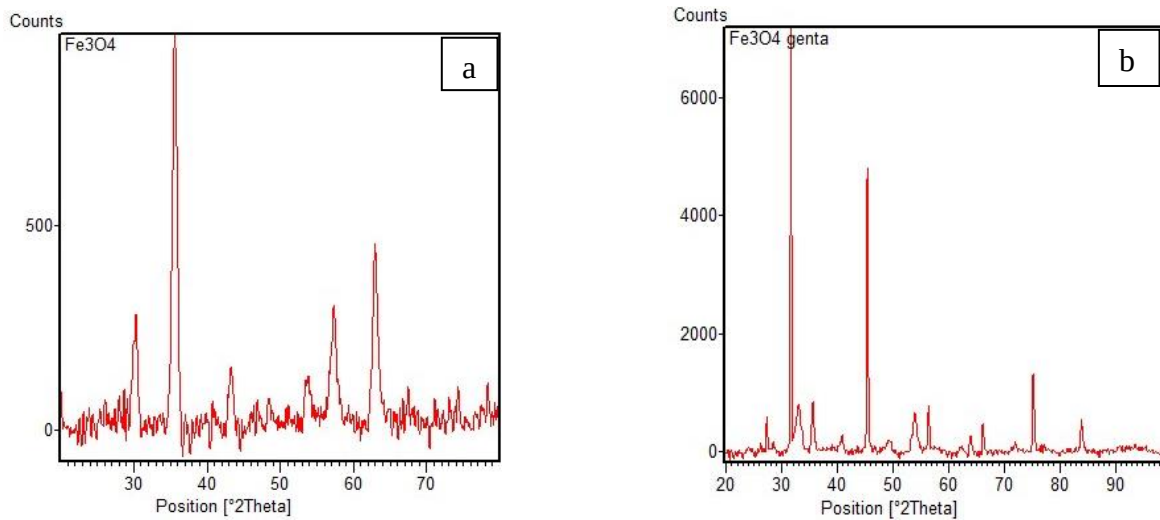


Figure 5 XRD study of the (a) uncoated magnetite nanoparticles and (b) Iron oxide nanoparticles coated with Gentamicin

3.1.4 Superconducting Quantum Interference Device (SQUID) Results:

Figure 6(a) shows the ZFC vs FC curve for the coated particles, obtained at a magnetic field of 100 Oe. From this curve, it is seen that the ZFC and FC curves separate out at the temperature of 300K (blocking temperature T_B).

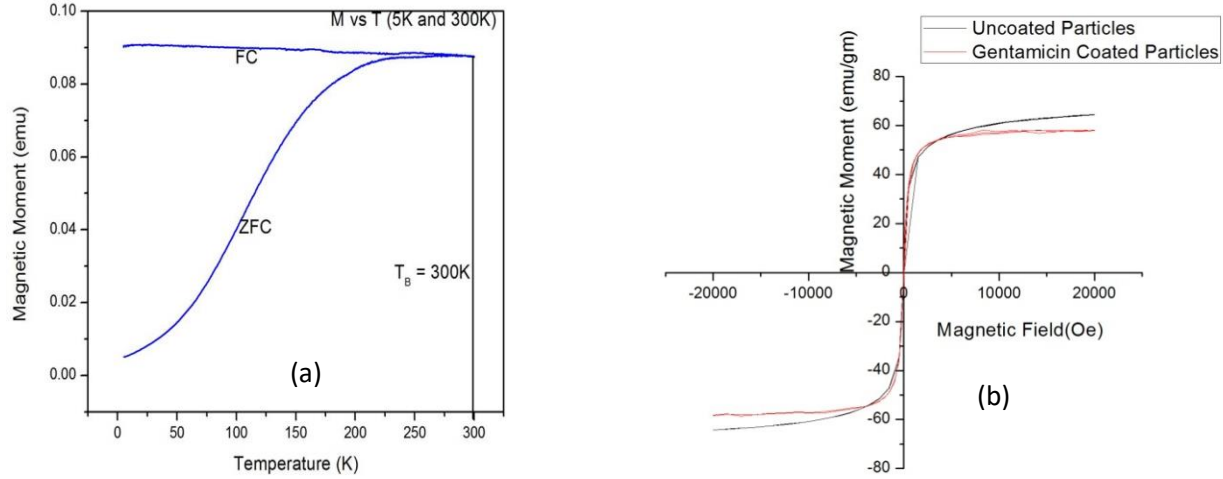


Figure 6 (a) ZFC vs FC curve for the coated nanoparticles showing the blocking temperature and (b) Magnetization Curve for the Nanoparticles

The magnetization curves of the synthesized nanoparticles (both uncoated and coated) are given in figure 6(b), obtained by keeping the temperature constant at 300K. The critical size that determines the super paramagnetic behavior is given by the formula:

$$D_s = \frac{k_B T}{K}$$

Where D_s is the critical diameter of the particles, k_B is the Boltsmann's constant ($1.38 \times 10^{-23} \text{ m}^2 \text{ kg/s}^2 \text{ K}$), K is the anisotropy constant (for Fe_3O_4 particles, it is $0.44 \times 10^4 \text{ J/m}^3$) and T is the absolute temperature.

Taking, $T = 300\text{K}$, we get $D_s = 28.65 \text{ nm}$.

Here, both the samples are less than 28.65 nm in diameter, and, therefore, they show super paramagnetic behavior. From the nature of the ZFC vs FC curves, we can conclude that the synthesized particles are super-paramagnetic at room temperature. The complete reversible nature of the hysteresis magnetization curve for both the coated and uncoated samples proves the super-magnetic behavior of the particles. The saturation magnetization value for the pristine and

antibiotic functionalized particles were found out to be about 66 emu/gm and 54 emu/gm respectively. The decrease in the M_s Value indicated the proper functionalization.

3.1.4 Zeta potential

The change in zeta potential with pH of the solution for the both types of nanoparticles is shown in figure 7. The iso-electric point (pH at which zeta potential is zero) for the uncoated particles came out to be 3 while that of coated particles came out to be 6.5. It may be noted that the higher negative potential of *E. coli* and *P.aeruginosa* than that of *S. aureas* and *E. coli* is due to the presence of an additional layer of negatively charged lipopolysaccharides in gram-negative bacteria than gram-positive bacteria.

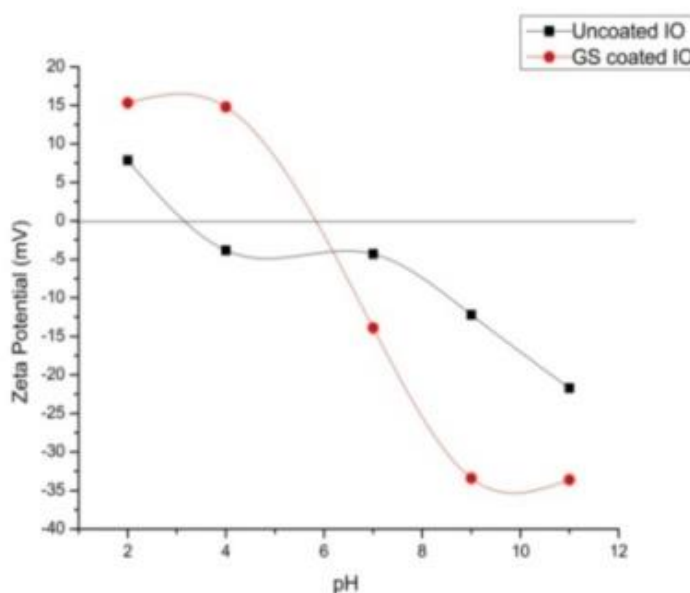


Figure 7 Variation in Zeta Potential with pH

Bacterial Strain	Zeta Potential(mV)	Zeta Potential(mV)
	Before treatment	After treatment
<i>S.aureus</i>	-30.2	-38.3
<i>E.coli</i>	-40.8	-46.2
<i>B.subtilis</i>	-20.9	-25.9
<i>P.aeruginosa</i>	-30.3	-37.8

Table 1 Zeta Potential for All Strains of Bacteria

After treatment with nanoparticles, the zeta potential of the bacterial strains was altered to an extent. The zeta potential values of the untreated and treated (with MIC values of the drug-nanoparticle conjugate) bacteria are detailed in table 1.

3.2 *In vitro Release of drug from the nanoparticles*

The release of Gentamicin from the iron oxide nanoparticles with time was analyzed and it showed a typical substantial release at beginning, gradually decreasing over time (as shown in figure 8). This is attributed to initial burst effect, owing to the high surface to volume ratio of the nanoparticles and weak bond formation of the drug with the nanoparticles. The burst release effect is very suitable in cases where the drug needs to be administered at varying rates, like the treatment of wounds or in several pulsatile delivery processes. Morphology of the host particle (size and presence of pores) and the type of host-drug interaction are considered two of the most important factors for burst release (37).

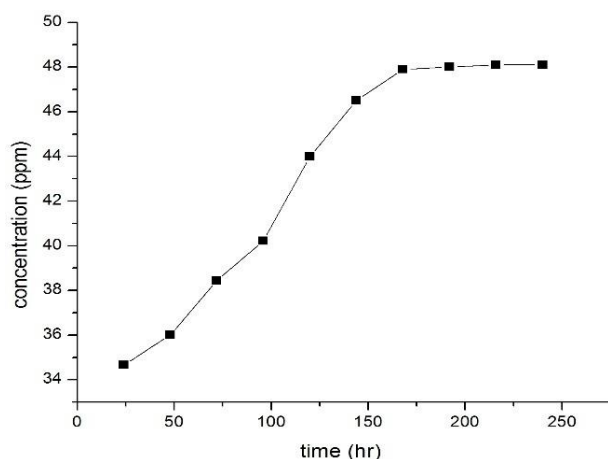


Figure 8 Release profiles of Gentamicin from nanoparticles

3.3 Zone of inhibition

The drug loaded nanoparticles showed high antimicrobial properties as confirmed by the zone of inhibition tests. The results are expressed in terms of increase in the area of the zone created by the nanoparticles, shown in figure 9, with an error bar of 4% approximately. All the experiments are conducted in triplicates. The well diffusion zone of inhibition area are more than disc diffusion ones.

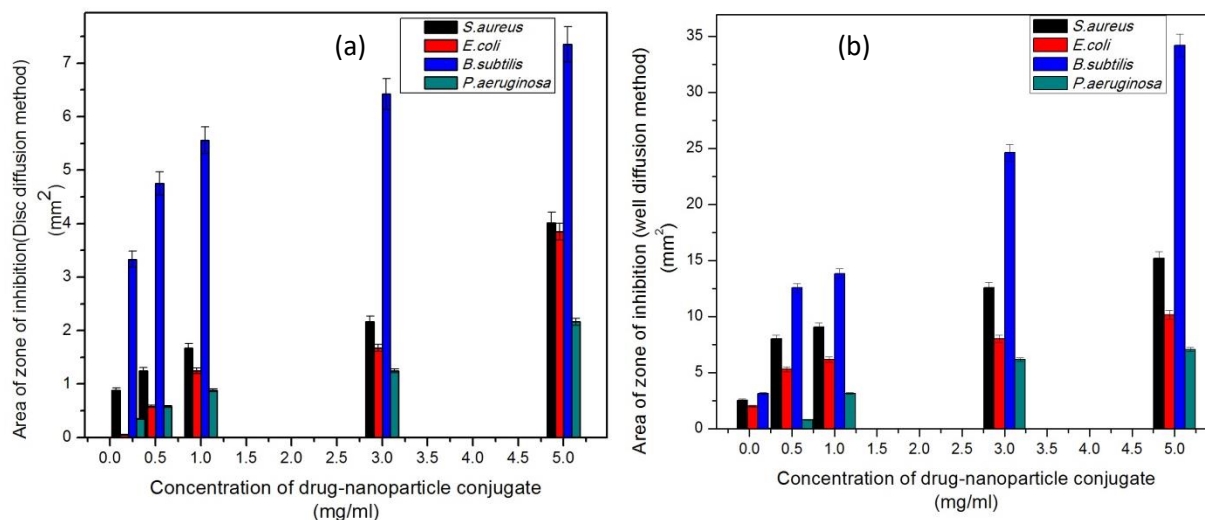


Figure 9 The area of the zone of inhibition created by nanoparticles in (a) disc and (b) well diffusion test

3.4 Colony forming Unit Study

Both the nanoparticles also showed great activity in terms of reduction of colony forming units.

The initial CFU for the control strains of bacteria were calculated as the following:

43.33×10^9 CFU/ml for *S.aureus*, 2.6×10^9 CFU/ml for *E.coli*, 3.5×10^{11} CFU/ml for *B.subtilis* and 2.94×10^{11} CFU/ml for *P.aeruginosa*. The reduction of CFU count of the bacterial strains with increasing concentration of the nanoparticles is given in figure 10 with an approximate error of 3%. It is evident from the plots shown in figure 10 that 98.45% reduction of the CFU count is done by 1.2mg/ml for uncoated particles whereas it requires only 0.2mg/ml of the coated particles for a reduction of >99%. Therefore, much smaller amount of GS coated iron oxide nanoparticles showed superior antibacterial action.

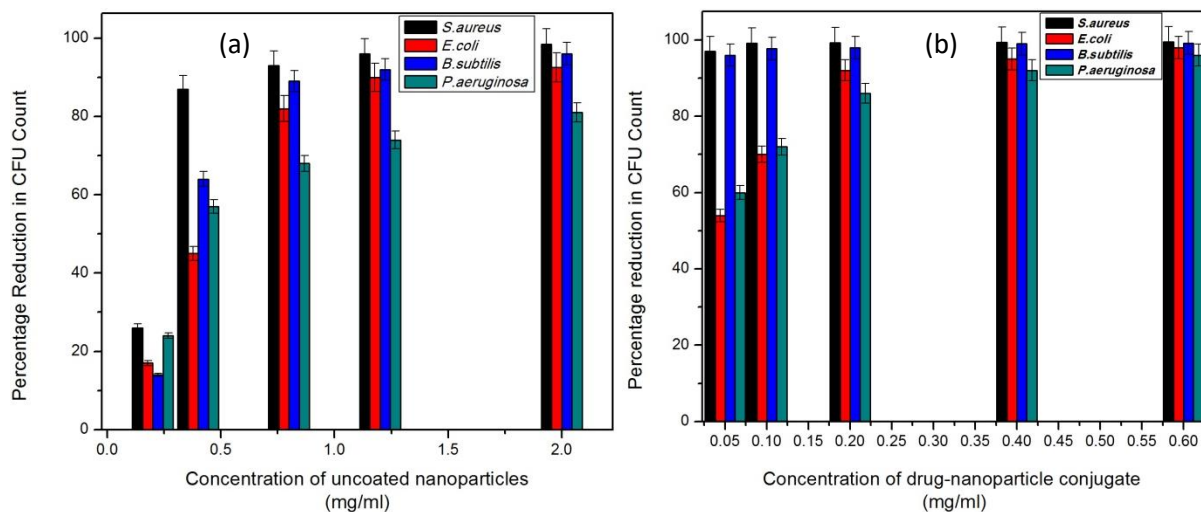


Figure 10 Comparison of the CFU reduction percentage as a function of concentration of (a) unloaded and (b) drug loaded nanoparticles

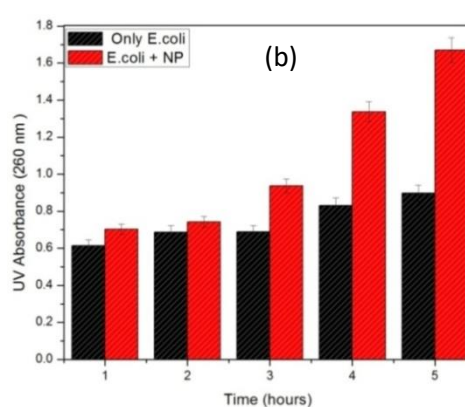
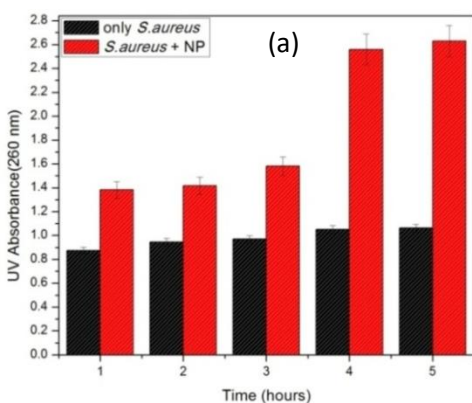
The MIC and MBC of the bacteria were also calculated for both the nanoparticles and is shown in table 2. The number of colonies appearing is more in case of *E. coli* and *P. aeruginosa* than in the case of *S. aureus* and *B. subtilis* which is in coherence with the Zone of Inhibition results that gram negative bacteria are more resistant than gram positive bacteria.

	<i>S.aureus</i>		<i>E.coli</i>		<i>B.subtilis</i>		<i>P.aeruginosa</i>	
	MIC (mg/ml)	MBC (mg/ml)	MIC (mg/ml)	MBC (mg/ml)	MIC (mg/ml)	MBC (mg/ml)	MIC (mg/ml)	MBC (mg/ml)
Uncoated NP	25	30	28	35	20	25	30	35
Coated NP	2	3	2.5	3	1.5	2.5	3	4

Table 2 MIC and MBC values for coated and uncoated NP for *S.aureus*, *B.subtilis*, *E.coli* and *P.aeruginosa*

3.5 Amino Acid Leakage

The amino acid leakage is usually measured in terms of absorbance. The absorbance of the solution is directly influenced by the concentration of the nucleic acids present in the solution. The wavelength of 260nm is chosen because nucleic acids show maximum absorbance at this wavelength. From the column graphs shown in figure 11(with an error bar of 3.5%), it can be interpreted that the increasing rate of absorbance for the nanoparticles treated bacterial solutions is an indication of increasing concentration of nucleic acids in the solution. It can be seen from the DNA leakage studies that the absorbance of the untreated bacteria remains almost constant, which proves that the nanoparticles are the only reason for DNA leakage.



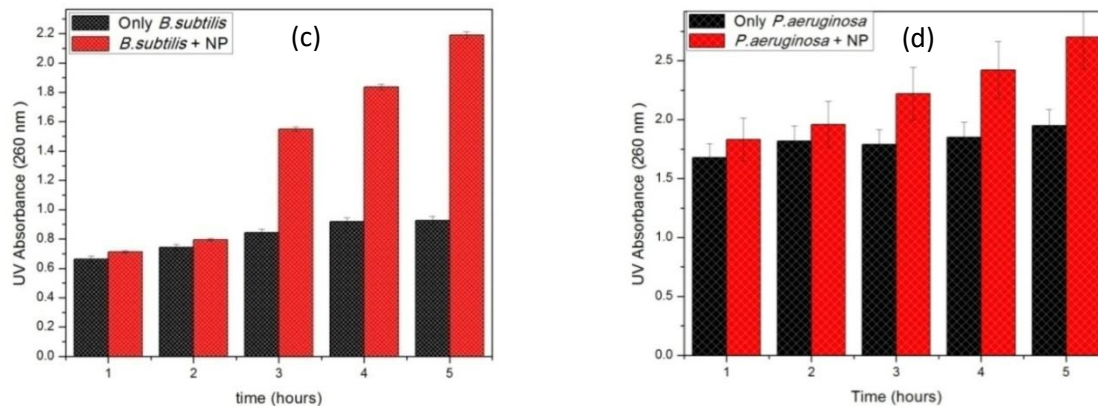


Figure 11 Increase in the UV absorbance at a wavelength of 260 nm of (a) *S.aureus*, (b) *E.coli*, (c) *B.subtilis* and (d) *P.aeruginosa* with time

3.6 Haemotoxicity

The RBC lysis induced by the nanoparticle-drug conjugate was analyzed using the haemocompatibility test and it is quantified with comparison to the lysis induced by 1% Triton-X(100% haemolysis) and PBS buffer(0% haemolysis). The acceptable limit for haemotoxicity is 5 %. The percent lysis is plotted against various concentrations of the conjugate and is shown in figure 12.

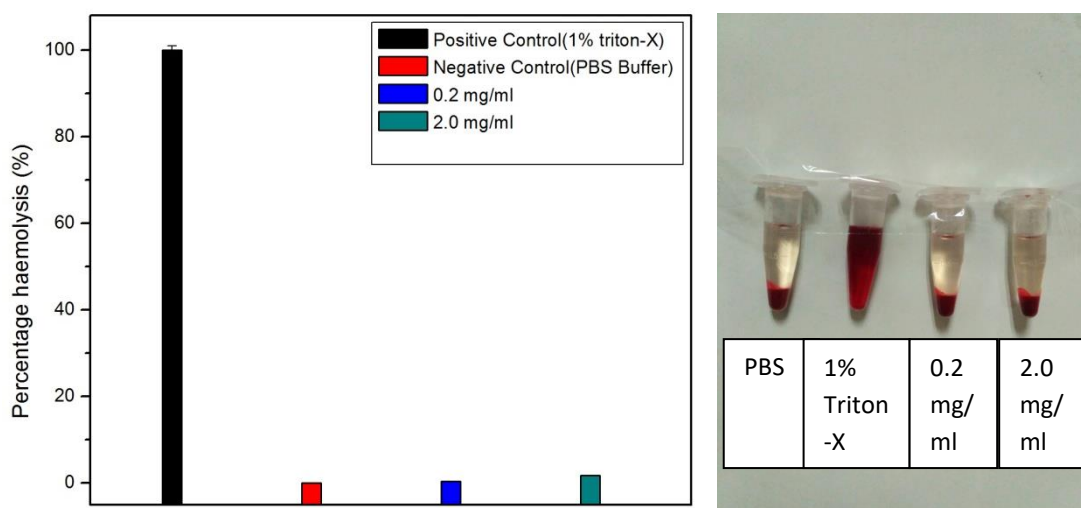


Figure 12 (a) Percentage haemolysis shown by different concentrations of drug-nanoparticle conjugate and (b) the original image of the positive, negative controls along with nanoparticles

It can be shown from figure 12 (a) that both 0.2 mg/ml (the dosage inhibiting ~98% growth of bacteria) and 2.0 mg/ml (MIC value) show negligible (0.43% and 1.7% respectively) haemolysis. Hence, it can be easily inferred that, the drug nanoparticle conjugate is practically compatible with human blood.

4. Discussions

Novel nanoparticles are very much in research due to its tunable properties and ability of fluorescence exhibition (38) which makes it a very eligible candidate for imaging of tumor cells. Drug loading on nanoparticles have gained attention recently and much work (tabulated in table 3) is going on in this arena. In this paper, Gentamicin sulphate is chosen because it is basically a complex made of closely related aminoglycosides. The main mechanism of action of this drug is the inhibition of protein biosynthesis or genetic translation. When conjugated with iron, Gentamicin forms phospholipids (39). It forms a layer around the nanoparticles, the sulphate group forming covalent bond with the iron (40). Then the conjugate takes part in the cell annihilation by mainly two mechanisms, namely, interruption of protein synthesis and inducing damage of cell membranes. Thus, it gives rise to a very effective drug carrier.

System	Dosage	Activity	Size and shape	Author
Carbapenems loaded gold nanoparticles	1-20 $\mu\text{g/ml}$	Formulation of conjugated gold nanoparticles enhances the antibacterial action of both imipenem and meropenem against carbapenem resistant bacteria.	35, 70 and 200 nm	Shaker et.al (19)

Biosynthesized Gold nanoparticle-based antibiotic conjugates	7-60 µg/ml	It showed significant annihilation towards multidrug-resistant isolates of coagulase-negative staphylococci.	12–32 nm	Roshmi et. al (21)
Paclitaxel-loaded biodegradable polymeric nanoparticles	1-5 mg	Drug release kinetics were adjusted by the inclusion of a co-surfactant, vitamin E TPGS or Poloxamer 188	440-600 nm	Sharma et. al (29)
Silver nanoparticles-ceftriaxone conjugate	1 mg/ml	MAGNPs–CEF were more effective against gram-negative and Gram-positive bacterial strains and were superior to ceftriaxone or free nanoparticles.	58 to 458 nm	Harshiny et. al (14)
Novel Nystatin Conjugated Iron Oxide Nanocomposite	2-5 mg/ml	Nyst-CS-MNP nanocomposite can be further used as an antifungal as well as antibacterial agent for human use with less toxicity.	6 to 23 nm	Hussein-Al-Ali et. al (20)
Ampicillin-conjugated magnetic nano-antibiotic	23.4 mg/ml	The synthesized nanocomposites exhibited antibacterial and antifungal properties, as well as antimycobacterial effects.	32.1 nm	Hussein-Al-Ali et. al (17)

Amikacin loaded PLGA nanoparticles	32 µg/ml	Nanoparticles showed no cytotoxicity against mouse macrophages RAW 264.7. Antibacterial and antibiofilm studies showed that the drug released was active against <i>P. aeruginosa</i> in vitro	247.5 ± 0.4 nm	Sabaeifard et. al (41)
Doxorubicin loaded magnetic chitosan nanoparticles	500 µg/ml	Doxorubicin loaded CS MNPs were absorbed by the cells and concentrated around the nucleus, so that drug is successfully delivered near to the nucleus, where it shows its anti-carcinogenic activity.	6–8 nm	Unsoy et. al (42)

Table 3 A comprehensive report of the available literatures in drug-nanoparticle conjugate

5. Conclusion

Iron Oxide nanoparticles were synthesized using the method of co-precipitation owing to its ease of operation, easily available raw materials, ability to process at room temperature and yield of pure products. This method also demonstrated its efficacy in the preparation of iron oxide NPs coated by a drug, Gentamicin. From the HR-TEM, SEM, and XRD analysis, the sizes of the uncoated and coated nanoparticles were calculated to be around 12nm and 9nm respectively. FTIR study showed the peaks characteristic of iron oxide and it shows extra peaks of amine groups after surface functionalization to affirm the proper surface impregnation by Gentamicin. These drug coated particles can be effectively used as a proper magnetic drug carrying vehicle.

The property of superparamagnetism, which means that there will be no residual magnetism after the removal of the external magnetic field, is affirmed by SQUID. This is the key for any particle to be used in-vivo because residual magnetism can create various problems within living cells. From the present study, it was shown that these particles have significant antibacterial efficacy different strains of gram positive and gram negative. The colony reduction of the strains proved this activity and it was found that a concentration of 2- 3mg/ml is sufficient to induce about 99% bacterial death, whereas a concentration as high as 30-35mg/ml is required for uncoated nanoparticles to inflict similar bacterial annihilation. Also, the disruption of the cellular DNA and RNA by the nanoparticles is confirmed by the amino acid leakage tests. Future studies on these particles can include the analysis of the haemo-toxicity of these particles to analyze their cell compatibility.

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